

# Assessment of Landfill Gas-to-Energy Potential and Technologies at Arefin Nagar and Fatehabad Landfills Using Site-Specific LandGEM Modeling

## ABSTRACT

The second-largest city in Bangladesh, Chattogram, produces over 3,000 tonnes of municipal solid waste (MSW) every day. The majority of this trash is dumped in open dumpsites at Arefin Nagar and Anandabazar without leachate treatment or gas control systems. Since organic waste makes up 62-73% of the MSW stream, anaerobic decomposition produces large amounts of landfill gas (LFG) that are now released uncontrollably into the atmosphere, posing both greenhouse gas and public health risks to nearby communities. As Chattogram City Corporation prepares to transition disposal operations to a new engineered landfill at Fatehabad by 2027, a critical opportunity exists to integrate gas-to-energy infrastructure into both the closing and incoming sites.

This study projects methane generation at Arefin Nagar and Fatehabad through 2090 using the USEPA LandGEM v3.02 Site Specific-1 (SP-1) model calibrated with locally determined parameters. According to modeling data, peak methane generation was 19,647 Mg yr<sup>-1</sup> at Arefin Nagar (2027) and 22,600 Mg yr<sup>-1</sup> at Fatehabad (2038). From 2030 to 2041, the overall generation exceeded 20,000 Mg yr<sup>-1</sup>. Converting this LFG to electricity via reciprocating IC engines yields peak recoverable outputs of 71.7 GWh/yr and 82.5 GWh/yr at the two sites, respectively, rising to over 130 GWh/yr under combined heat-and-power (CHP) operation at higher collection efficiency.

Based on these findings, LFG recovery via reciprocating IC engines is identified as the most technically and economically suitable near-term energy extraction technique for Chattogram, owing to its compatibility with high-moisture organic waste, modular low-cost deployment (USD 1,000-2,500/kW), and ability to capture methane already accumulated within existing waste mass. The waste stream's high moisture content and low calorific value (3-5 MJ/kg) make incineration, pyrolysis, and gasification inappropriate. Anaerobic digestion is recommended as a complementary medium-term pathway for fresh organic waste, subject to the development of source-separation infrastructure. A phased implementation strategy is proposed, beginning with LFG well-field deployment at Arefin Nagar ahead of its 2026-2027 closure.

*Keywords: LandGEM; landfill gas; methane recovery; waste-to-energy; anaerobic digestion; municipal solid waste*

## 1. INTRODUCTION

Chattogram, the second-most populous city in Bangladesh and gateway to its international trade, faces an acute MSW crisis, generating approximately 3,000 tonnes per day, of which roughly 2,200 tonnes are directed to the overloaded Arefin Nagar (Bayezid) and Anandabazar (Halishahar) open dumpsites. Neither site is equipped with engineered liners, leachate treatment, or gas-control systems, resulting in uncontrolled LFG release and continuous leachate seepage into the Karnaphuli River [1].

Methane, the principal energy-bearing component of LFG, is approximately 81.2 times more potent than CO<sub>2</sub> as a greenhouse gas over a 20-year horizon, and landfills rank as the third-largest anthropogenic methane source globally [15]. With roughly 75% of Chattogram's MSW comprising organic or food waste, anaerobic decomposition generates substantial, currently uncaptured LFG, and communities near Arefin Nagar report elevated rates of respiratory illness, ... ..

*Will be uploaded after the presentation in December 2026*